Active ants: analysis

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- Well-posedness

- Mean-field limit: large stochastic system $\Longleftrightarrow$ deterministic PDE

- Lipschitz and moment-controlled ∇K and $\sigma_x \geq 0$: solutions exist up to any time and the empirical measure converges to the mean-field limit PDE

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- Well-posedness
  - Singular (Newtonian) ∇K and $\sigma_x > 0$: $|\nabla K(x)| \sim |x|^{-1}$